Supplementary Material for "Voltage-Only FAS-MPC for Standalone LC-Filtered Inverters under Nonlinear Loads"

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This document provides the derivations, simulation-model and baseline details, tuning protocol, and extended numerical results referenced in the manuscript. Equation, table, figure, and reference numbers without the prefix "S" refer to the manuscript.

S1 Derivations

FAS form. From $C_f\dot{v}_c = i_f - i_\ell$ with $y = v_c$, $i_f = C_f\dot{y} + i_\ell$. Differentiating and substituting $L_f\dot{i}_f = u_{\text{inv}} - y - r_f i_f$ gives

$$C_f\ddot{y} + \tfrac{\mathrm{d}i_\ell}{\mathrm{d}t} = \tfrac{1}{L_f}\big(u_{\text{inv}} - y - r_f C_f\dot{y} - r_f i_\ell\big), \tag{S1}$$

which is (2) after division by C_f. For a nominal load $i_{\ell,0}(y) = G_0 y + \frac{2}{3}P_0 y S^{-1}(y)$, $\frac{\mathrm{d}}{\mathrm{d}t}i_{\ell,0} = J_{\ell,0}(y)\dot{y}$ with $J_{\ell,0}(y) = G_0 I_2 + \frac{2}{3}P_0\big(S(y)I_2 - 2yy^{\mathrm{T}}\big)S^{-2}(y)$. The design uses $P_0 = 0$, so that $\hat{\Phi}(z) = -K_\Phi z$ with $K_\Phi = [a_0 I_2,\ a_1 I_2]$, $a_0 = (1 + r_0 G_0)/(L_0 C_0)$, $a_1 = r_0/L_0 + G_0/C_0$. A nominal CPL term makes $\hat{\Phi}$ nonlinear, so the QP then needs a drift approximation, e.g., freezing $\hat{\Phi}$ along the previous prediction. For a fixed set of weights, $P_0 = 600$ W with this approximation and $P_0 = 0$ gave RMS errors of 3.55/3.76/4.63 V and 3.59/3.53/4.67 V on the default plant and the corners $(0.7, 0.9)$ and $(1.3, 1.1)$, respectively; $P_0 = 0$ is therefore used.

Exact discretization. With $A_c = \left[\begin{smallmatrix} 0 & I_2 \\ 0 & 0 \end{smallmatrix}\right]$, $A_c^2 = 0$, hence $e^{A_c\tau} = \sum_{j\ge 0}(A_c\tau)^j/j! = I_4 + A_c\tau$. Therefore

$$A_d = I_4 + A_c T_s = \begin{bmatrix} I_2 & T_s I_2 \\ 0_{2\times 2} & I_2 \end{bmatrix}, \qquad B_d = \int_0^{T_s} \big(I_4 + A_c(T_s - \tau)\big)\begin{bmatrix} 0 \\ I_2 \end{bmatrix}\mathrm{d}\tau = \begin{bmatrix} \int_0^{T_s}(T_s - \tau)\mathrm{d}\tau\, I_2 \\ T_s I_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}T_s^2 I_2 \\ T_s I_2 \end{bmatrix}. \tag{S2}$$

The lower-left block of A_d is $0_{2\times 2}$, i.e., $\dot{y}_{k+1} = \dot{y}_k + T_s\mu_k$ and $y_{k+1} = y_k + T_s\dot{y}_k + \frac{1}{2}T_s^2\mu_k$. Model (S2) is exact for the virtual input held over the interval. In the inverter, the voltage computed at t_k is held; the resulting intersample deviation is part of $d_{r,k}$ and w_k in (5), and the stability test of Section S5 discretizes the physical LC plant with held voltage exactly, so it contains this effect without approximation. The implementation uses these closed forms; they coincide with `scipy.linalg.expm` of the augmented matrix $\left[\begin{smallmatrix} A_c & B_c \\ 0 & 0 \end{smallmatrix}\right] T_s$ to machine precision (zero difference for $T_s = 50$ μs).

DOB with compensated prediction. With (6) of the manuscript, $\epsilon_k = \hat{z}_k - A_d\hat{z}_{k-1} - B_d(\mu_{k-1} + \hat{d}_{r,k-1})$. With the posterior-residual estimate $\bar{d}_{r,k} = \hat{d}_{r,k-1} + B_d^{\dagger}\epsilon_k$, the update can equivalently be written as the first-order filter $\hat{d}_{r,k} = (1-\alpha)\hat{d}_{r,k-1} + \alpha\bar{d}_{r,k}$, which equals the integral update $\hat{d}_{r,k} = \hat{d}_{r,k-1} + \alpha B_d^{\dagger}\epsilon_k$, i.e., the integral mode of (7). Including $\hat{d}_{r,k-1}$ in the KF prediction compensates for the estimated constant disturbance and makes the estimate unbiased for constant disturbances (internal model principle); omitting this term would leave a steady-state bias proportional to the KF velocity error.

CPL Lipschitz bounds. With $P_0 = 0$, the CPL contributes $\psi = -\frac{2P}{3}\big[\frac{r_f}{L_f C_f}g(y) + \frac{1}{C_f}J_g(y)\dot{y}\big]$, $g = yS^{-1}$, $J_g = (SI_2 - 2yy^{\mathrm{T}})S^{-2}$, with the *true* L_f, C_f, r_f; the nominal drift contains no CPL term, so no cancellation between true and nominal CPL terms is assumed. On the tube $\mathcal{D}$ of Section IV, $S \ge V_{\min}^2 + \varepsilon_v$, and the eigenvalues of J_g are $1/S$ and $(\varepsilon_v - \|y\|^2)/S^2$, so $\|J_g\| \le 1/(V_{\min}^2 + \varepsilon_v)$, which gives ℓ_v. For the y-derivative of $J_g(y)v$, $\partial_y(J_g v) = -2vy^{\mathrm{T}}S^{-2} - 2\big((y^{\mathrm{T}}v)I_2 + yv^{\mathrm{T}}\big)S^{-2} + 8(y^{\mathrm{T}}v)yy^{\mathrm{T}}S^{-3}$. By rotation invariance, let $y = re_1$ and $v = [c, s]^{\mathrm{T}}$ with $c^2 + s^2 = 1$. Then $\partial_y(J_g v) = rS^{-2}M$ with $M = \left[\begin{smallmatrix} ac & -2s \\ -2s & -2c \end{smallmatrix}\right]$ and $a = 8r^2/S - 6 \le 2$. M is symmetric, and $\|M\| = \frac{1}{2}\big[(2-a)|c| + \sqrt{(a+2)^2c^2 + 16s^2}\big]$, whose maximum over c is $2\sqrt{8/(a+6)} = 2\sqrt{S}/r$ (Cauchy–Schwarz; attained at $c^2 = S/(4r^2) \le 1$, i.e., for $r^2 \ge \varepsilon_v/3$). Hence

$$\sup_{\|v\|=1}\|\partial_y(J_g(y)v)\| = 2\big(\|y\|^2 + \varepsilon_v\big)^{-3/2},$$

which decreases in $\|y\|$, so $2S_{\min}^{-3/2}$ enters ℓ_y (for $\varepsilon_v = 0$, it equals $2/\|y\|^3$). Dense sampling of $\|y\|$ and of the angle between y and v reproduces this value to 10^{-9} (Table S10). The tube cross-section is convex at each t, so the mean-value bounds hold between any two states compared at the same time.

Sample-held representation and tube check. In the discrete small-gain test, ψ is evaluated at t_k and held over each 50-μs sample; along the reference, it varies by at most $\omega T_s \approx 1.6\%$ per sample, but inter-sample variation (including the PWM ripple of $\dot{y}$) is not covered by the test. `tube_check.py` evaluates $E_y = \max_k \|y_k - v_{\mathrm{ref}}(t_k)\|/V_{\mathrm{pk}}$ and $E_v = \max_k \|\dot{y}_k - \dot{v}_{\mathrm{ref}}(t_k)\|/(\omega V_{\mathrm{pk}})$, with the true $\dot{y} = (i_f - i_\ell)/C_f$, at the sampling instants of the switching-level main test (five noise realizations; default plant and the four corners of Ω). In steady CPL operation (25–50 ms), $E_y \le 0.037$ and $E_v \le 1.95$ (0.029 and 0.81 at the default plant), inside the tube $\delta_y = 0.1V_{\mathrm{pk}}$, $\delta_v = 2\omega V_{\mathrm{pk}}$ of Section IV; this is an observation over the reported runs, not a proof that the tube is invariant. During the CPL step (20–25 ms), $E_y \le 0.21$ and $E_v \le 4.1$, so the step transient lies outside the tube and is assessed only by simulation. On the 5-μs record, which contains the PWM ripple, E_v reaches 2.6 in steady CPL operation. The derivative errors are dominated by measurement noise and ripple; tighter tubes (e.g., $\delta_v = 0.2\omega V_{\mathrm{pk}}$) are not satisfied in the simulations and are therefore not used.

S2 Simulation Models and Metrics

Power stage. States $i_f, v_c \in \mathbb{R}^2$; integration by semi-implicit Euler with $h = 0.1$ μs. The CPL draws $i = \frac{2}{3}P_{\mathrm{CPL}}v_c/(v_c^{\mathrm{T}}v_c + \varepsilon_v)$; the rectifier is a three-phase diode bridge with $L_s = 0.5$ mH, $R_s = 0.05$ Ω per phase, and $C_d = 470$ μF in parallel with $R_d = 45$ Ω, simulated as a hybrid system (conduction sets updated at each step; the floating common-mode potential follows from $\sum_x \mathrm{d}i_x/\mathrm{d}t = 0$; diodes turn off at current reversal; C_d precharged to $0.95\sqrt{3}V_{\mathrm{pk}}$).

Switching level. Leg references from inverse Clarke transformation with min–max zero-sequence injection (space-vector PWM, linear range $\|u\| \le V_{\mathrm{dc}}/\sqrt{3}$); duty cycles computed with the measured dc-link voltage; 10-kHz triangular carrier with update at peak and valley ($T_s = 50$ μs); one-sample computation delay; dead time $t_d = 1$ μs (both switches off; the freewheeling diode is selected by the current sign); on-state drop 1 V; dc-link ripple $\pm 1\%$ at 300 Hz; 12-bit ADC over ± 400 V plus Gaussian noise $\sigma_v = 0.5$ V; capacitor-current sensor noise 0.05 A (PR+AD only).

Averaged model. $u_{\mathrm{inv}} = u_{\mathrm{cmd}}$ held over T_s, no delay, no ADC quantization.

Metrics. $e(t) = v_c(t) - v_{\mathrm{ref}}(t)$ sampled at 200 kHz. RMS over $[0, 120)$ ms and windows Base $[5, 20)$, CPL $[20, 50)$, NL $[50, 80)$, Sag $[80, 100)$, Final $[100, 120)$ ms; Peak: $\max\|e\|$ in $[20, 30)$ ms; t_{rec}: last time in $[20, 50)$ ms at which $\|e\| > 0.05V_{\mathrm{pk}}$, measured from the CPL step (n.r.: not recovered); THD of v_α up to the 50th harmonic over $[60, 80)$ (NL) and $[100, 120)$ ms (final); Clip: percentage of samples in which the raw command exceeded $V_{\mathrm{dc},k}/\sqrt{3}$ by more than 0.1%, with $V_{\mathrm{dc},k}$ the sample used when the command was computed (the QP tolerance is about 10^{-6}); the largest ratio before projection and the ratio with respect to the dc-link sample at the application instant are also recorded (Table S10); Active: percentage of samples with an active polygon constraint at stage 0. $\pm$ values are standard deviations over ten noise realizations; all controllers use the same seeds and the same noise sequence per sample (the 40-kHz FCS-MPC draws more samples). Worst: maximum RMS over the grid $L_f/L_0 \in \{0.7, 0.85, 1, 1.15, 1.3\}$, $C_f/C_0 \in \{0.9, 1, 1.1\}$ (seed 0). "Default plant" denotes $L_f = 0.9L_0$, $C_f = 1.1C_0$, $r_f = 1.5r_0$.

QP implementation. With $P_0 = 0$, the stage constraints $c_j^{\mathrm{T}}u_{i|k} \le \bar{U}_k \cos(\pi/M)$ are affine in η with a constant matrix; at each sample, the unconstrained minimizer $-H^{-1}q_k$ is evaluated first and the OSQP solve (warm start, tolerance 10^{-6}, at most 4000 iterations) is skipped when it satisfies all NM inequalities. "Proposed, no QP" never calls the solver: it applies the unconstrained law and clips the command to the disk.

S3 Baseline Implementations

Selected baseline parameters: FAS-PD $K_p = \omega_c^2$, $K_d = 2\zeta\omega_c$, $\omega_c = 2\pi \cdot 1.5$ kHz, $\zeta = 0.5$; MA-LADRC $\omega_c = 2\pi \cdot 0.8$ kHz, $\omega_o = 2\pi \cdot 4$ kHz; PR+AD $K_c = 20$ Ω, $K_{pv} = 0.15$ S, $K_{r,h} = 5$ S; FCS-MPC weight 3/10 and observer poles $\{0.6, 0.65, 0.7\}/\{0.5, 0.55, 0.6\}$ at 40/20 kHz.

B3 FAS-PD. Same KF and delay compensation as the proposed controller; $\mu_k = \ddot{v}_{\mathrm{ref}} - \omega_c^2(\hat{y} - v_{\mathrm{ref}}) - 2\zeta\omega_c(\hat{\dot{y}} - \dot{v}_{\mathrm{ref}})$, followed by the FAS map and projection.

B4 MA-LADRC. Linear ADRC [10], [11] with the controller and observer bandwidths ω_c, ω_o of [11]; in addition, the known nominal LC terms are included in the observer model (model-assisted). Per axis, the extended state observer uses the exact discretization of $\ddot{y} = b_0 u - a_0 y - a_1 \dot{y} + f$, $\dot{f} = 0$, with $b_0 = 1/(L_0 C_0)$, $a_0 = (1 + r_0 G_0)/(L_0 C_0)$, $a_1 = r_0/L_0 + G_0/C_0$, in current-estimator form with a triple pole at $e^{-\omega_o T_s}$ (Ackermann). Control: $u = [\ddot{r} + \omega_c^2(r - \hat{y}) + 2\omega_c(\dot{r} - \widehat{\dot{y}}) + a_0\hat{y} + a_1\widehat{\dot{y}} - \hat{f}]/b_0$ with one-step prediction for the delay, followed by projection. A plain LADRC with $a_0 = a_1 = 0$ was also tested; it produced a steady error above 16 V because the large resonant term $y/(L_0C_0)$ enters the total disturbance, so the model-assisted version is used.

B5 PR+AD. $i_c^\star = C_0 \dot{v}_{\mathrm{ref}} + K_{pv} e_v + \sum_{h\in\{1,5,7\}} R_h(s) e_v$ with $R_h(s) = 2K_{r,h}\omega_b s/(s^2 + 2\omega_b s + h^2\omega^2)$ discretized exactly (zero-order hold); $u = v_c + K_c(i_c^\star - i_c)$ (capacitor-voltage feedforward and capacitor-current active damping); measured v_c and i_c.

B6 FCS-MPC. As in [5], only v_c is measured and the currents needed for the prediction are estimated. Instead of the capacitor-current estimator of [5], a Luenberger observer (current-estimator form) of (i_f, v_c, i_ℓ) per axis with i_ℓ modeled as constant is used; one-step delay compensation; enumeration of the eight switching states at $k+2$ with cost $\|v_{\mathrm{ref}} - v_c\|^2 + \lambda \left\| C_0 \dot{v}_{\mathrm{ref}} - (i_f - \hat{i}_\ell) \right\|^2$ plus a 10^{-6} tie-break on switching transitions; dead time applied at every transition.

S4 Tuning Protocol

Each controller was tuned on the switching-level model (seed 0) by grid search, minimizing $J_{\mathrm{rob}} = \frac{1}{2}(\mathrm{RMS}_{\mathrm{def}} + \max_c \mathrm{RMS}_c)$ over the default plant and the corners $L_f/L_0 \in \{0.7, 1.3\}$, $C_f/C_0 \in \{0.9, 1.1\}$ (five runs per candidate). Grids: proposed $\sigma_a \in \{1,3\}\sigma_a^\circ$ ($\sigma_a^\circ = 3.84 \times 10^8$ V/s^2), $\alpha \in \{0.1, 0.2, 0.35\}$, $r_\eta \in \{3\times10^{-7}, 3\times10^{-6}\}$, $q_v \in \{0.01, 0.05\}$, $N \in \{5, 10\}$, $(\mathcal{H}, \beta_h) \in \{(\emptyset, 0), (\{1,-5,7\}, 0.2/0.5/1), (\{1,-5,7,-11,13\}, 0.5)\}$; FAS-PD $\sigma_a \in \{0.3,1,3\}\sigma_a^\circ$, $\omega_c/2\pi \in \{0.8, 1, 1.5, 2, 2.5\}$ kHz, $\zeta \in \{0.5, 0.7, 1\}$; MA-LADRC $\omega_c/2\pi \in \{0.8, \ldots, 2.5\}$ kHz, $\omega_o/2\pi \in \{2, \ldots, 6\}$ kHz; PR+AD $K_c \in \{5, \ldots, 25\}$ Ω, $K_{pv} \in \{0.05, \ldots, 0.2\}$ S, $(K_{r,1}, K_{r,5/7}) \in \{(5,5), (10,5), (5,2)\}$ S; FCS-MPC $\lambda \in \{0, 0.3, 1, 3\}$ (40 kHz) or $\{0.3, 1, 3, 10\}$ (20 kHz) and three or four observer-pole sets. Selected values: proposed $J_{\mathrm{rob}} = 4.04$ (default 3.42, worst corner 4.67 V); PR+AD 3.29 (2.70, 3.89); FCS-MPC 40 kHz 3.72 (3.24, 4.21); FAS-PD 6.03 (5.41, 6.65); MA-LADRC 6.06 (5.14, 6.98); FCS-MPC 20 kHz 14.56 (8.88, 20.24); about 3500 tuning runs in total. During the grid search, the constraints of the FAS-based controllers were assembled with the drift frozen along the previous prediction, which is cheaper to build; the exact linear-drift constraints (9), which are used for all reported results, change the default-plant RMS by less than 0.02 V. With a nominal-only criterion, all voltage-only designs (including the proposed one) were tuned close to their stability limit and became unstable at $L_f/L_0 = 0.7$; the robust criterion was therefore adopted for every controller. B1 is the best $\mathcal{H} = \emptyset$ design of the same grid ($\alpha = 0.35$, $r_\eta = 3\times10^{-6}$, $q_v = 0.01$, $N = 10$), and B2 reuses the proposed weights without the DOB; hence B1 (Table I) differs from the sensitivity point $\beta_h = 0$ (Fig. S1), which keeps all other proposed weights.

S5 Linear Closed Loop for the Stability Test

For linear loads with conductance G and an inactive limit, the loop is linear in error coordinates with the state

$$\xi_k = \left[x_k^{\mathrm{T}}, \hat{z}_{k-1}^{\mathrm{T}}, \hat{d}_{k-1}^{0\mathrm{T}}, \hat{d}_{k-1}^{1\mathrm{T}}, \hat{d}_{k-1}^{-5\mathrm{T}}, \hat{d}_{k-1}^{7\mathrm{T}}, \eta_{0|k-1}^{\mathrm{T}}, \mu_{k-1}^{\mathrm{T}}, \mu_k^{\mathrm{T}}, u_k^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{R}^{24},$$

where $x_k = [y_k^{\mathrm{T}}, \dot{y}_k^{\mathrm{T}}]^{\mathrm{T}}$ is the sampled plant state of (1), and μ_k and u_k are the virtual input and the command applied over $[t_k, t_{k+1})$, both computed at t_{k-1}. One sample of the loop reads

$$\begin{aligned}
x_{k+1} &= F x_k + g_u u_k + g_d d_{r,k},\\
\hat{z}_{k|k-1} &= A_d \hat{z}_{k-1} + B_d(\mu_{k-1} + \hat{d}_{r,k-1}), \qquad \hat{z}_k = \hat{z}_{k|k-1} + K_\infty(H x_k + \nu_k - H\hat{z}_{k|k-1}),\\
&\quad \text{DOB (7) driven by } \epsilon_k = \hat{z}_k - \hat{z}_{k|k-1},\\
z_{0|k} &= A_d \hat{z}_k + B_d(\mu_k + \hat{d}_{r,k}), \qquad \eta_{0|k} = -K_z z_{0|k} + K_\eta \eta_{0|k-1},\\
\mu_{k+1} &= \eta_{0|k} - \hat{d}_{1|k}, \qquad u_{k+1} = L_0 C_0 \big(\mu_{k+1} + K_\Phi z_{0|k}\big).
\end{aligned}$$

Here, (F, g_u, g_d) is the exact zero-order-hold discretization of (1) with held voltage and held matched disturbance; K_∞ is the stationary gain of the discrete Riccati equation; K_z and K_η are the stage-0 rows of the unconstrained minimizer $-H_{\mathrm{qp}}^{-1} q_k$, whose linear term q_k is linear in $(z_{0|k}, \eta_{0|k-1})$; and $\hat{d}_{1|k}$ is the one-step preview in (7). Every map is linear, so $\xi_{k+1} = A_{\mathrm{cl}}\xi_k + E\varpi_k$. Without delay, $z_{0|k} = \hat{z}_k$ and the last two blocks of ξ_k are dropped ($\xi_k \in \mathbb{R}^{20}$).

Time-varying Kalman gain. The implemented KF runs the Riccati recursion from $P_0 = \mathrm{diag}(\sigma_v^2 I_2, (0.05\omega V_{\mathrm{pk}})^2 I_2)$. Its gain satisfies $\|K_k - K_\infty\| / \|K_\infty\| < 10^{-2}$ for $k \geq 3$ and $< 10^{-6}$ for $k \geq 5$, i.e., after 0.25 ms. The implemented loop is thus $\xi_{k+1} = (A_{\mathrm{cl}} + \Delta_k)\xi_k + (E + \Delta_k^E)\varpi_k$ with $\Delta_k, \Delta_k^E \to 0$ exponentially. Because A_{cl} is Schur, a quadratic Lyapunov function of A_{cl} decreases along this time-varying loop for all k beyond a finite k_0, so the implemented loop is exponentially stable and input-to-state stable as well.

Execution order. With delay compensation, each sample executes: (1) sample $y_{m,k}$ and $V_{\mathrm{dc},k}$; (2) KF update (6) with μ_{k-1} (the input applied over $[t_{k-1}, t_k)$) and $\hat{d}_{r,k-1}$; (3) DOB update (7); (4) prediction $z_{0|k} = A_d \hat{z}_k + B_d(\mu_k + \hat{d}_{r,k})$, where μ_k was computed at t_{k-1} and is applied over $[t_k, t_{k+1})$; (5) QP with references and previews shifted by one sample and $\bar{U}_k$ from $V_{\mathrm{dc},k}$; (6) μ_{k+1} and u_{cmd} from (3) at $z_{0|k}$, applied over $[t_{k+1}, t_{k+2})$. A_{cl} and E are obtained by applying the linear update above to unit vectors. The reference enters only through exogenous terms (its least-squares residual and the drift mismatch along v_{ref}, a matched fundamental-frequency term).

S6 Extended Results

Table S1: Switching-level metrics (mean ± standard deviation over ten noise realizations)

Controller	RMS V	Base V	CPL V	NL V	Sag V	Final V	Peak V	t_{rec} ms	$\mathrm{THD}_{\mathrm{NL}}$ %	$\mathrm{THD}_{\mathrm{fin}}$ %	Clip %	Active %
Proposed	3.43±0.04	1.64	2.72	4.55	3.66	3.43	23.34	0.56	2.09	2.08	0.00	1.70
Proposed, no QP	3.39±0.04	1.64	2.64	4.55	3.53	3.42	23.23	0.48	2.10	2.08	1.58	1.73
B1: w/o harmonic models	5.77±0.02	0.91	2.10	7.39	7.31	7.51	21.93	0.38	4.82	4.79	0.00	1.46
B2: w/o DOB	23.13±0.02	14.85	20.29	26.23	25.54	26.51	40.97	n.r.	5.22	5.28	0.00	0.02
B3: FAS-PD	5.42±0.01	2.44	3.60	6.56	6.52	6.62	24.33	0.36	3.43	3.44	0.52	0.00
B4: MA-LADRC	5.17±0.04	1.43	2.34	6.59	6.46	6.55	21.89	0.37	4.23	4.20	2.66	0.00
B5: PR+AD	2.69±0.01	1.39	1.46	3.67	2.90	2.84	16.71	0.28	1.99	1.80	0.13	0.00
B6: FCS-MPC (20 kHz)	8.67±0.33	6.49	6.86	9.96	9.38	9.89	20.27	n.r.	6.01	5.65	–	–
B6: FCS-MPC (40 kHz)	3.30±0.07	2.16	2.23	3.83	3.98	3.89	12.49	0.19	2.33	2.43	–	–

Table S2: Averaged-model metrics (mean over five noise realizations)

Controller	RMS V	Base V	CPL V	NL V	Sag V	Final V	Peak V	t_{rec} ms	$\mathrm{THD}_{\mathrm{NL}}$ %	$\mathrm{THD}_{\mathrm{fin}}$ %	Clip %	Active %
Proposed	2.47	0.53	1.62	3.57	2.68	2.46	15.43	0.32	1.91	1.54	0.00	0.82
B1: w/o harmonic models	3.82	0.55	1.31	4.94	4.85	4.94	15.27	0.28	3.21	3.18	0.00	0.76
B2: w/o DOB	11.33	1.66	6.73	13.97	14.27	14.27	16.98	1.07	4.97	4.97	0.00	0.04
B3: FAS-PD	3.56	0.55	1.55	4.54	4.54	4.53	15.32	0.26	2.67	2.68	0.15	0.00
B4: MA-LADRC	3.39	0.96	1.54	4.36	4.22	4.25	15.27	0.28	2.75	2.68	1.17	0.00
B5: PR+AD	2.39	0.73	0.94	3.44	2.70	2.61	9.27	0.21	1.85	1.67	0.04	0.00

Table S3: One-at-a-time sensitivity of the proposed controller (switching level, three realizations; $\beta_h = 0$ means $\mathcal{H} = \emptyset$)

Parameter	Tested values → RMS error (V)
N	3→3.79; 5→3.52; 10→3.43; 20→3.45
α	0.05→3.79; 0.1→3.43; 0.2→3.76; 0.35→7.87
β_h	0→7.69; 0.2→3.64; 0.5→3.43; 1→3.26
σ_a (V/s^2)	1.15e+08→4.48; 3.45e+08→3.66; 1.15e+09→3.43; 3.45e+09→4.51
r_η	3e-08→3.39; 3e-07→3.40; 3e-06→3.43; 3e-05→3.68
q_v	0.01→5.49; 0.05→3.43; 0.2→4.55; 1→7.98
κ_f	0→3.56; 1→3.51; 5→3.43; 20→3.63
γ_u	0.9→4.52; 0.95→3.72; 1→3.43
M	8→3.73; 12→3.49; 24→3.43; 48→3.41

Table S4: Robustness tests (RMS error in V; noise, dead time, and CPL: two realizations; sag: three realizations). FAS-PD under the sag: 5.41, 5.43, 5.53 V.

Test	Value	Proposed	MA-LADRC	PR+AD	FCS-MPC (40 kHz)
σ_v	0.2 V	3.33	5.22	2.69	3.25
σ_v	0.5 V	3.43	5.16	2.70	3.29
σ_v	1 V	4.47	5.54	2.73	3.86
σ_v	2 V	12.44	12.07	2.86	6.89
t_d	0 μs	3.15	5.18	2.62	3.44
t_d	0.5 μs	3.27	5.17	2.65	3.29
t_d	1 μs	3.43	5.16	2.70	3.29
t_d	2 μs	3.87	5.15	2.82	3.34
P_{CPL}	0.6 kW	3.22	5.01	2.60	3.29
P_{CPL}	1.2 kW	3.43	5.16	2.70	3.29
P_{CPL}	1.8 kW	4.02	5.48	3.01	3.49
P_{CPL}	2.4 kW	5.53	6.38	3.93	3.78
Sag depth	340 V	3.43	5.18	2.69	3.30
Sag depth	320 V	3.76	5.19	2.71	3.43
Sag depth	300 V	4.87	5.61	3.32	3.68

Table S5: RMS error (V) over the parameter grid: Proposed (bold: unbounded/unstable operation, RMS $>$ 10 V)

$L_f/L_0 \setminus C_f/C_0$	0.8	0.9	1	1.1	1.2
0.6	**118.7**	**86.0**	7.1	3.2	2.9
0.7	**88.0**	4.7	3.3	3.1	3.0
0.85	4.1	3.5	3.4	3.3	3.3
1	3.8	3.7	3.6	3.6	3.6
1.15	4.1	4.0	4.0	3.9	3.9
1.3	4.4	4.3	4.3	4.3	4.3
1.4	4.6	4.6	4.6	4.6	4.6

Table S6: RMS error (V) over the parameter grid: B3: FAS-PD (bold: unbounded/unstable operation, RMS $>$ 10 V)

$L_f/L_0 \setminus C_f/C_0$	0.8	0.9	1	1.1	1.2
0.6	**76.0**	**15.2**	6.3	4.6	4.4
0.7	**14.3**	5.5	4.8	4.7	4.7
0.85	5.5	5.3	5.2	5.2	5.2
1	5.8	5.8	5.8	5.8	5.8
1.15	6.2	6.2	6.2	6.2	6.2
1.3	6.7	6.6	6.6	6.6	6.6
1.4	6.9	6.9	6.9	6.9	6.9

Table S7: RMS error (V) over the parameter grid: B4: MA-LADRC (bold: unbounded/unstable operation, RMS $>$ 10 V)

$L_f/L_0 \setminus C_f/C_0$	0.8	0.9	1	1.1	1.2
0.6	**89.4**	**79.8**	**11.5**	4.2	3.8
0.7	**81.0**	6.9	4.4	4.3	4.2
0.85	5.5	4.9	4.9	4.9	4.9
1	5.5	5.5	5.6	5.6	5.7
1.15	6.1	6.2	6.2	6.3	6.4
1.3	6.7	6.8	6.9	7.0	7.1
1.4	7.1	7.2	7.3	7.4	7.5

Table S8: RMS error (V) over the parameter grid: B5: PR+AD (bold: unbounded/unstable operation, RMS $>$ 10 V)

$L_f/L_0 \setminus C_f/C_0$	0.8	0.9	1	1.1	1.2
0.6	8.8	6.5	4.6	4.1	3.3
0.7	4.6	2.8	2.3	2.3	2.2
0.85	2.9	2.7	2.6	2.6	2.6
1	3.2	3.1	3.0	2.9	2.9
1.15	3.6	3.5	3.4	3.3	3.3
1.3	4.0	3.9	3.8	3.7	3.6
1.4	4.3	4.2	4.0	4.0	3.9

Table S9: RMS error (V) over the parameter grid: B6: FCS-MPC (40 kHz) (bold: unbounded/unstable operation, RMS > 10 V)

$L_f/L_0 \setminus C_f/C_0$	0.8	0.9	1	1.1	1.2
0.6	**18.1**	6.8	4.3	3.2	3.1
0.7	5.3	3.8	3.4	3.1	3.0
0.85	3.5	3.6	3.4	3.2	3.2
1	3.7	3.6	3.4	3.6	3.7
1.15	4.1	3.8	3.8	3.9	3.9
1.3	4.3	4.2	4.3	4.1	4.4
1.4	4.5	4.6	4.4	4.5	4.7

Table S10: Stability test, small-gain test, QP check, computation time, and long runs (CPU: Intel Core i9-14900, single process)

Quantity	Value
Dense grid: points per case, all Schur stable	6405, True
Largest sampled $\rho(A_{\mathrm{cl}})$ with / without delay compensation	0.99965 / 0.99968
Input-gain range preserving stability (default plant)	[0.01, 1.55]
Extra delay causing instability	2 samples
Gain to y at $h = 0$: matched sequence / full 2×2	3.1e-23 / 1.1e-22
Gain to y at $h = 1$: matched sequence / full 2×2	1.8e-21 / 2.2e-08
Gain to y at $h = -5$: matched sequence / full 2×2	7.7e-22 / 7.7e-08
Gain to y at $h = 7$: matched sequence / full 2×2	4.7e-22 / 2.4e-08
ℓ_2 gains γ_y, γ_v (default plant)	1.70e-07, 1.58e-04
$V_{\min}^3 \sup_{\|v\|=1} \|\partial_y(J_g v)\|$ for $\delta_y = 0.2V_{\mathrm{pk}}$: closed form / sampled	1.954024 / 1.954024
Sampled E_y / E_v, steady CPL, 25–50 ms: all five plants (default plant)	0.037 / 1.95 (0.029 / 0.81)
Sampled E_y / E_v, CPL step, 20–25 ms: all five plants (default plant)	0.212 / 4.11 (0.152 / 3.38)
Small-gain index, 600 W, $(\delta_y, \delta_v) = (0.1V_{\mathrm{pk}}, 2\omega V_{\mathrm{pk}})$: default / worst plant	0.32 / 0.45
Small-gain index, 1200 W, $(\delta_y, \delta_v) = (0.1V_{\mathrm{pk}}, 2\omega V_{\mathrm{pk}})$: default / worst plant	0.64 / 0.91
Small-gain index, 1800 W, $(\delta_y, \delta_v) = (0.1V_{\mathrm{pk}}, 2\omega V_{\mathrm{pk}})$: default / worst plant	0.96 / 1.36
Small-gain index, 2400 W, $(\delta_y, \delta_v) = (0.1V_{\mathrm{pk}}, 2\omega V_{\mathrm{pk}})$: default / worst plant	1.28 / 1.81
Small-gain index, 1200 W, $(\delta_y, \delta_v) = (0.2V_{\mathrm{pk}}, 2\omega V_{\mathrm{pk}})$: default / worst plant	0.88 / 1.24
QP vs. SOCP (8): samples (active polygon), failures	448 (166), 0
$\|\Delta u_0\|$ mean / 95th pct. / max	0.55 / 2.23 / 7.07 V
Largest predicted QP command over the horizon / $\bar{U}_k$	1.0000000
QP time, 340-V sag (median / 99th / max, C level)	82 / 529 / 1108 μs
Samples with feasible unconstrained law, 340-V sag	98.3%
QP time, 300-V sag (median / 99th / max, C level)	82 / 832 / 1298 μs
Samples with feasible unconstrained law, 300-V sag	93.2%
1-s run, Proposed: RMS / THD of last cycle	3.31 V / 2.10%
1-s run, B1: w/o harmonic models: RMS / THD of last cycle	7.59 V / 4.83%
1-s run, B4: MA-LADRC: RMS / THD of last cycle	6.66 V / 4.14%

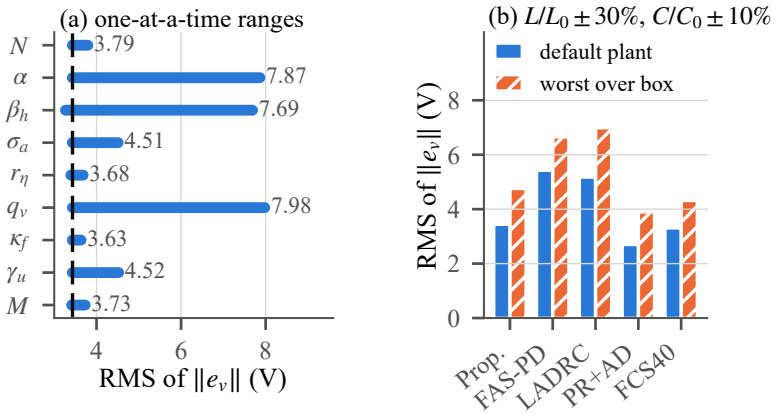


Figure S1: (a) RMS-error range of the proposed controller when each parameter is varied alone; ticks mark the values of Table II, and $\beta_h = 0$ keeps the other weights, unlike the separately tuned B1. (b) Default-plant and worst-case errors of all controllers.

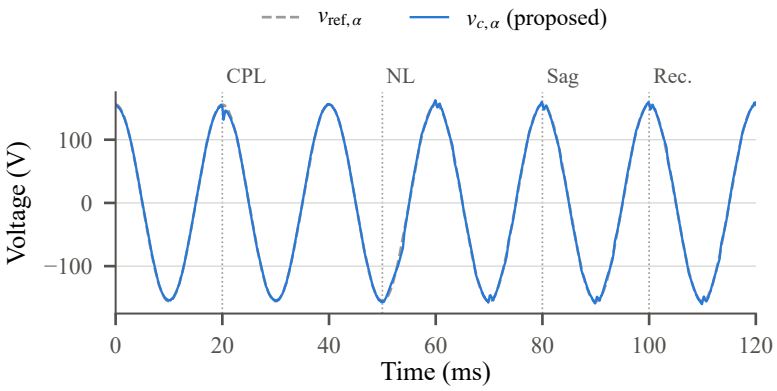


Figure S2: Phase-α capacitor voltage of the proposed controller (switching level) and the reference.

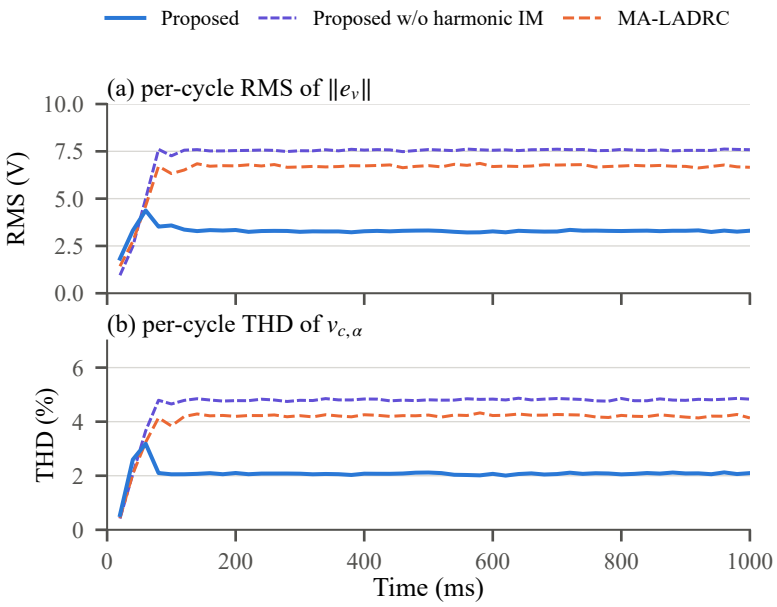


Figure S3: One-second runs (same events as the main test, then steady operation with all loads and $V_{\mathrm{dc}} = 400$ V): per-cycle RMS of $\|e_v\|$ and THD. The error of the proposed controller is stationary after about 0.2 s, so the slow harmonic-estimate modes cause no drift.

S7 Reproducibility

The Python code (`plant.py`: Numba plant models; `sim.py`: controllers, runner, metrics; `robust_tune.py`: tuning protocol; `experiments.py`: all reported runs; `analysis.py`, `box_cert.py`, `small_gain.py`, `tube_check.py`: stability test, small-gain test, tube check, and QP check; `lc_extra.py`, `lc_fb.py`: worst-case grids; `figures.py`, `supp_tables.py`: figures and tables) reproduces every number of the manuscript and of this document from fixed random seeds.